Chapter 2

Defining Data's Role in AI

IN THIS CHAPTER

- **Seeing data as a universal resource**
- **Obtaining and manipulating data**
- **Looking for mistruths in data**
- **Defining data-acquisitions limits**
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There is nothing unique about data. Every interesting application ever written for a computer has data associated with it. Data comes in many forms — some organized, some not. What has changed is the *amount* of data. Some people find it almost terrifying that access to so much data that details nearly every aspect of their lives is a reality — sometimes to a level that even the terrified people don't realize. In addition, the use of advanced hardware and improvements in algorithms make data now *the* universal resource for AI.

To work with data, you must first obtain it. Today, data is collected manually, as done in the past, and also automatically, using new methods. However, it's not a matter of just one or two data collection techniques: Collection methods take place on a continuum from fully manual to fully automatic. You also find a focus today on collecting this data ethically — for example, not collecting data that a person hasn't granted permission for. This chapter explores issues surrounding data collection.

Raw data doesn't usually work well for analysis purposes. This chapter also helps you understand the need for manipulating and shaping the data so that it meets specific requirements. You also discover the need to define the truth value of the data to ensure that analysis outcomes match the goals set for applications in the first place.



REMEMBER Interestingly, you also have data-acquisition limits to deal with. No technology currently exists for grabbing thoughts from someone's mind by telepathic means. Of course, other limits exist, too — most of which you probably already know about but may not have considered. It also doesn't pay to collect data in a manner that isn't secure. The data must be free of bias, uncorrupted, and from a source you know. You find out more about acquisition limits and data security in this chapter.

Finding Data Ubiquitous in This Age

You may have heard big data mentioned in many specialized scientific and business publications, and you may have even wondered what the term really means. From a technical perspective, *big data* refers to large and complex amounts of computer data, so large and intricate that applications can't deal with the data by simply using additional storage or increasing computer power.

Big data implies a revolution in data storage and manipulation. It affects what you can achieve with data in more qualitative terms (meaning that in addition to doing more, you can perform tasks better). From a human perspective, computers store big data in different data formats (such as database files and .csv files), but regardless of storage type, the computer still sees data as a stream of ones and zeros (the core language of computers). You can view data as being one of two types, structured and unstructured, depending on how you produce and consume it. Some data has a clear structure (you know exactly what it contains and where to find every piece of data), whereas other data is unstructured (you have an idea of what it contains, but you don't know exactly how it is arranged).

- **Structured data** include typical examples such as database tables, in which information is arranged into columns and each column contains a specific type of information. Data is often structured by design. You gather it selectively and record it in its correct place. For example, you may want to place a count of the number of people buying a certain product in a specific column, in a specific table, or in a specific database. As with a library, if you know what data you need, you can find it immediately.



- TIP **Unstructured data** consists of images, videos, and sound recordings. You may use an unstructured form for text so that you can tag it with characteristics, such as size, date, or content type. Usually, you don't know exactly where data appears in an unstructured dataset, because the data appears as sequences of ones and zeros that an application must interpret or visualize.



REMEMBER Transforming unstructured data into a structured form can cost lots of time and effort and can involve the work of many people. Most of the data of the big data revolution is unstructured and stored as is, unless someone renders it structured.

This copious and sophisticated data store didn't appear suddenly overnight. It took time to develop the technology to store this amount of data. In addition, it took time to spread the technology that generates and delivers data — namely, computers, sensors, smart mobile phones, and the internet and its web services. The following sections help you understand what makes data a universal resource today.

Using data everywhere

Scientists need more powerful computers than the average person because of their scientific experiments. They began dealing with impressive amounts of data years before anyone coined the term *big data.* At that point, the internet wasn't producing the vast sums of data that it does today.



REMEMBER Big data isn't a fad created by software and hardware vendors but has a basis in many scientific fields, such as astronomy (space missions), satellite (surveillance and monitoring), meteorology (storm predictions), physics (particle accelerators), and genomics (DNA sequences).

Although an AI application can specialize in a scientific field — such as IBM's Watson, which boasts an impressive medical-diagnosis capability because it can learn information from millions of scientific papers on diseases and medicine — the actual AI application driver often has more mundane facets. Actual AI applications are mostly prized for being able to recognize objects, move along paths, or understand what people say and speak to them. Data contribution to the actual AI renaissance that molded it in such a fashion didn't derive from the classical sources of scientific data.



TIP The internet now generates and distributes new data in large amounts. Our current daily data production is estimated to amount to about 2.5 quintillion (a number with 18 zeros) bytes, with the lion's share going to unstructured data like video and audio.

All this data is related to common human activities, feelings, experiences, and relations. Roaming through this data, an AI can easily learn how reasoning and acting more human-like works. Here are some examples of the more interesting data you can find:

- **Large repositories of faces and expressions from photos and videos posted on social media websites like Facebook, YouTube, and Google:** They provide information about gender, age, feelings, and possibly sexual orientation, political orientation, or IQ (see "Face-reading AI will be able to detect your politics and IQ, professor says" at The `Guardian.com`).
- **Privately held medical information and biometric data from smartwatches, which measure body data such as temperature and heart rate during both illness and good health:** Interestingly enough, data from smartwatches is seen as a method to detect serious diseases, such as COVID-19, early.

- **Datasets of how people relate to each other and what drives their interest from sources such as social media and search engines:** For instance, a study from Cambridge University's Psychometrics Centre claims that Facebook interactions contain a lot of data about intimate relationships.
- **Information on how we speak, which is recorded by mobile phones.** For example, OK Google, a function found on Android mobile phones, routinely records questions and sometimes even more, as explained in "Google's been quietly recording your voice; here's how to listen to — and delete — the archive" at `https://qz.com/526545/googles-been-quietly-recording-your-voice-heres-how-to-listen-to-and-delete-the-archive`.

Every day, users connect even more devices to the internet that start storing new personal data. There are now personal assistants that sit in houses, such as Amazon Echo and other integrated smart home devices that offer ways to regulate and facilitate the domestic environment. These are just the tip of the iceberg because many other common tools of everyday life are becoming interconnected (from the refrigerator to the toothbrush) and able to process, record, and transmit data. The internet of things (IoT) is becoming a reality.

Putting algorithms into action

The human race is now at an incredible apex of unprecedented volumes of data, generated by increasingly smaller and powerful hardware. The data is also increasingly processed and analyzed by the same computers that the process helped spread and develop. This statement may seem obvious, but data has become so ubiquitous that its value no longer resides only in the information it contains (such as the case of data stored in a firm's database that allows its daily operations), but rather in its use as a means to create new values. Some people call such data the "new oil." These new values exist mostly in how applications manicure, store, and retrieve data, and in how you actually use it by means of smart algorithms.

Algorithms and AI changed the data game. As mentioned in Chapter 1, AI algorithms have tried various approaches along the way in the following order:

1. Simple algorithms

2. Symbolic reasoning based on logic
3. Expert systems

In recent years, AI algorithms have moved to neural networks and, in their most mature form, deep learning. As this methodological passage happened, data turned from being the information processed by predetermined algorithms to becoming what molded the algorithm into something useful for the task. Data turned from being just the raw material that fueled the solution to the artisan of the solution itself, as shown in Figure 2-1.

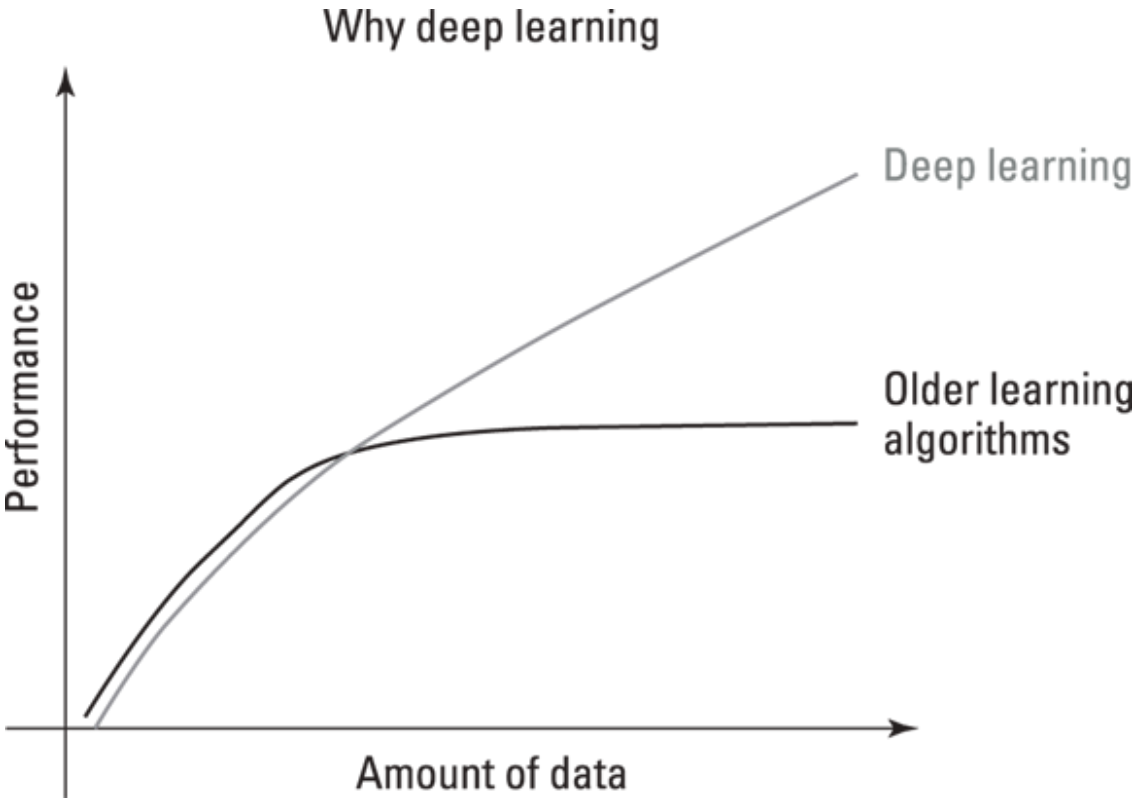


FIGURE 2-1: With the present AI solutions, more data equates to more intelligence.

Thus, a photo of your kittens has become increasingly useful not simply because of its affective value — depicting your cute little cats — but also because it could become part of the learning process of an AI discovering more general concepts, such as which characteristics denote a cat or understanding what defines cute.

On a larger scale, a company like Google feeds its algorithms from freely available data, such as the content of websites or the text found in publicly available texts and books. Google Spider software crawls the web, jumping from website to website and retrieving web pages with their

content of text and images. Even if Google gives back part of the data to users as search results, it extracts other kinds of information from the data using its AI algorithms, which learn from it how to achieve other objectives.

Algorithms that process words can help Google AI systems understand and anticipate your needs even when you're expressing them not in a set of keywords but in plain, unclear natural language, the language we speak every day (and yes, everyday language is often unclear). If you currently try to pose questions, not just chains of keywords, to the Google search engine, you'll notice that it tends to answer correctly. Since 2012, with the introduction of the Hummingbird update, Google has steadily become better able to understand synonyms and concepts, a strategy that goes beyond the initial data it acquired, and this is the result of an AI process.

A few years after Hummingbird, Google deployed an even more advanced algorithm named RankBrain, which learns directly from millions of queries every day and can answer ambiguous or unclear search queries, even expressed in slang or colloquial terms, or simply riddled with errors. RankBrain doesn't service all the queries, but it learns from data how to better answer queries.

Using Data Successfully

Having plentiful data available isn't enough to create a successful AI. Presently, an AI algorithm can't extract information directly from raw data. Most algorithms rely on external collection and manipulation prior to analysis. When an algorithm collects useful information, it may not represent the right information. The following sections help you understand how to collect, manipulate, and automate data collection from an overview perspective.

Considering the data sources

The data you use comes from a number of sources. The most common data source is from information entered by humans at some point. Even

when a system collects shopping-site data automatically, humans initially enter the information. A human clicks various items, adds them to a shopping cart, specifies characteristics (such as size and quantity), and then checks out. Later, after the sale, the human gives the shopping experience, product, and delivery method a rating and makes comments. In short, every shopping experience becomes a data collection exercise as well.



O'REILLY

REMEMBER Many data sources today rely on input gathered from human sources. Humans also provide manual input. You call or go into an office somewhere to make an appointment with a professional. A receptionist then gathers information from you that's needed for the appointment. This manually collected data eventually ends up in a dataset somewhere for analysis purposes.

Data is also collected from sensors, and these sensors can take almost any form. For example, many organizations base physical data collection, such as the number of people viewing an object in a window, on cell-phone detection. Facial recognition software could potentially detect repeat customers.

However, sensors can create datasets from almost anything. The weather service relies on datasets created by sensors that monitor environmental conditions such as rain, temperature, humidity, and cloud cover.



TIP Robotic monitoring systems help correct small flaws in robotic operation by constantly analyzing data collected by monitoring sensors. A sensor, combined with a small AI application, could tell you when your dinner is cooked to perfection tonight. The sensor collects data, but the AI application uses rules to help define when the food is properly cooked.

Obtaining reliable data

The word *reliable* seems so easy to define, yet so hard to implement. Something is reliable when the results it produces are both expected and consistent. A reliable data source produces mundane data that contains no surprises; no one is shocked in the least by the outcome. Depending on your perspective, it could actually be a good thing that most people aren't yawning and then falling asleep when reviewing data. The surprises make the data worth analyzing and reviewing. Consequently, data has an aspect of duality. We want reliable, mundane, fully anticipated data that simply confirms what we already know, but the unexpected is what makes collecting the data useful in the first place.

Still, you don't want data that is so far out of the ordinary that it becomes almost frightening to review. Balance needs to be maintained when obtaining data. The data must fit within certain limits (as described in the "Manicuring the Data" section, later in this chapter). It must also meet the specific criteria of truth value (as described in the "Considering the Five Mistruths in Data" section, later in this chapter). The data must also come at expected intervals, and all the fields of the incoming data record must be complete.



REMEMBER To some extent, data security also affects data reliability. Data consistency comes in several forms. When the data arrives, you can ensure that it falls within expected ranges and appears in a particular form. However, after you store the data, the reliability can decrease unless you ensure that the data remains in the expected form. An entity fiddling with the data affects reliability, making the data suspect and potentially unusable for analysis later. Ensuring data reliability means that after the data arrives, no one tampers with it to make it fit within an expected domain (making it mundane as a result).

Making human input more reliable

Humans make mistakes — it's part of being human. In fact, expecting that humans won't make mistakes is unreasonable. Yet many application designs assume that humans somehow won't make mistakes of any sort.

The design expects that everyone will simply follow the rules. Unfortunately, the vast majority of users are guaranteed to not even read the rules because most humans are also lazy or too pressed for time when it comes to doing things that don't really help them directly.

Consider the entry of a state into a form. If you provide just a text field in the application gathering the data, you encounter a wide variety of possible input:

- Some users might input the entire state name, such as Kansas. Of course, some users will make a typo or a capitalization error and come up with Kanzuz, Kansus, or kANSAS. You see the potential for errors.
- People and organizations also have various approaches to performing tasks. Someone in the publishing industry might use the Associated Press (AP) style guide and input Kan. Someone who is older and used to the Government Printing Office (GPO) guidelines might input Kans. instead.
- And some people might go for other available state abbreviations. The U.S. Post Office (USPS) uses KS, but the U.S. Coast Guard uses KA. Meanwhile, the International Standards Organization (ISO) form goes with US-KS.

Mind you, this is just a state entry, which is reasonably straightforward — or so you thought before reading this section. Clearly, because the state won't change names anytime soon, you could simply provide a drop-down list box on the form for choosing the state in the required format, thereby eliminating differences in abbreviation use, typos, and capitalization errors in one fell swoop.



REMEMBER Drop-down list boxes work well for an amazing array of data inputs, and using them ensures that human input into those fields becomes extremely reliable because the human has no choice but to use one of the default entries. Of course, the human can always choose the incorrect entry, which is where double-checks come into play. Some newer applications compare the zip code to the city and state entries to see whether they match. When they don't match (sometimes it's just a matter of capitalization), the user is asked again to provide the correct input. This double-check verges on being annoying, but the user is unlikely to see it often, so it shouldn't become too annoying.

Even with cross-checks and static entries, humans still have plenty of room for making mistakes. For example, entering numbers can be problematic. When a user needs to enter 2.00, you may see 2, or 2.0, or 2., or any of a variety of other entries. Fortunately, parsing the entry and reformatting it fixes the problem, and you can perform this task automatically, without the user's aid. (Unfortunately, some online sites want you to enter information like credit cards with dashes, some with spaces, and some with no spacing at all, which makes for a confusing session when the application doesn't fix the entry automatically.)

Unfortunately, reformatting doesn't correct an errant numeric input. You can partially mitigate such errors by including range checks — a customer can't buy fewer than zero bars of soap. And, unless the customer is filthy or owns a wombat farm, entering 50,000 bars of soap would likely be a mistake, too. The legitimate way to show that the customer is returning the five bars of soap is to process a return, not a sale. However, the user may have simply made an error, and you can provide a message stating the proper input range for the value.

Using automated data collection

Some people think that automated data collection solves all the human-input issues associated with datasets. In fact, automated data collection does provide a number of benefits:

- Better consistency
- Improved reliability
- Lower probability of missing data
- Enhanced accuracy
- Reduced variance for factors like timed inputs

Unfortunately, to say that automated data collection solves every issue is simply incorrect. Automated data collection still relies on sensors, applications, and computer hardware designed by humans that provide access only to the data that humans decide to allow. Because of the limits that humans place on the characteristics of automated data collection, the outcome often provides less helpful information than the designers

hoped for. Consequently, automated data collection is in a constant state of flux as designers try to solve the input issues.



REMEMBER Automated data collection also suffers from both software and hardware errors present in any computing system, but with a higher potential for *soft issues* (which arise when the system is apparently working but isn't providing the desired result) than other kinds of computer-based setups. When the system works, the reliability of the input far exceeds human abilities. However, when soft issues occur, the system often fails to recognize, as a human might, that a problem exists, and therefore, the dataset could end up containing more mediocre or even bad data.

Collecting data ethically

For some people, anything that appears on the internet is automatically considered public domain — including people's faces and all their personal information. The fact is that you should consider everything as being copyrighted and unavailable for use in a public domain manner to use data safely. Even people who realize that material is copyrighted often fall back on fair use principles. Fair use can be quite a tricky subject, as witnessed by the *Author's Guild v. Google* case. The case centered on Google's Google Books project, in which Google created a searchable database of printed copyrighted books. The case was finally decided in favor of Google, but only because Google had met some strict requirements. In addition, this kind of fair use is about books, not people.

The problem with considering fair use alone is that it's also essential to consider a person's right to privacy (you can read about various laws in "Internet privacy laws revealed — how your personal information is protected online" at `legal.thomsonreuters.com`). Consequently, it shouldn't surprise anyone that a major ruckus arose when companies started scraping (screen scraping) images of people wearing masks from the internet without obtaining any permission whatsoever. In fact, Facebook sued and lost a lawsuit over its misuse of user data.

The right to privacy has also created a new industry for making a person's face less useful to companies that are determined to get free data without permission by using any means possible (see the *New York Times* article "This Tool Could Protect Your Photos from Facial Recognition"). The fact is, no matter where you stand on the free-use issue, you still need to consider the ethical use of data that you obtain, no matter what the source may be. Here are some considerations to keep in mind as you collect personal data ethically:

- **Obtaining permission:** Some research requires you to be able to identify persons used within a dataset. Going out and grabbing *p*ersonally *i*dentifiable *i*nformation (PII) isn't a good way to gather data. For one thing, you can't be sure that the information is either complete or correct, so any analysis you perform is suspect. For another thing, you could encounter the messy and costly consequences of legal actions. The best way to obtain data with PII is to ask permission.
- **Using sanitization techniques:** *Data sanitization* involves removing personal information — such as name, address, telephone number, and ID — from a dataset so that identifying a particular individual in a dataset becomes impossible. In addition to text and dataset variables, you must consider every kind of data. For example, if you're working with collections of photos, it is *paramount* that you take steps to blur faces and remove car plates from images.
- **Avoiding data inference:** When collecting data, some users refuse to share personally identifiable information, such as gender and age. Some people recommend you infer this information when a user's picture or other information is available. Unfortunately, names associated with one gender in a particular culture may be assigned to another gender in other cultures. The problem with age inference is even more profound. For example, a machine learning algorithm will likely infer the wrong age for a person with albinism, which can affect as many as 1 in 3,000 individuals, depending on the part of the world the data comes from.
- **Avoiding generalizations:** Many fields of study try to incorrectly apply statistics and machine learning outcomes, with the result that an individual ends up being mistreated in some manner.



REMEMBER It's essential to remember that statistics apply to groups, not to individuals.

Manicuring the Data

Some people use the term *manipulation* when speaking about data, giving the impression that the data is somehow changed in an unscrupulous or devious manner. Perhaps a better term is *manicuring,* which makes the data well-shaped and lovely. No matter which term you use, however, raw data seldom meets the requirements for processing and analysis. To get something from the data, you must manicure it to meet specific needs. The following sections discuss data manicuring needs.

Dealing with missing data

To answer a given question correctly, you must have all the facts. You can guess the answer to a question without all the facts, but then the answer is just as likely to be wrong as correct. Often, someone who makes a decision, essentially answering a question, without all the facts is said to jump to a conclusion. When analyzing data, you have probably jumped to more conclusions than you think because of missing data. A *data record,* or one entry in a *dataset* (which is all the data), consists of *fields* that contain facts used to answer a question. Each field contains a single kind of data that addresses a single fact. If that field is empty, you don't have the data you need to answer the question using that particular data record.



REMEMBER As part of the process of dealing with missing data, you must know that the data is missing. Identifying that your dataset is missing information can be quite difficult because it requires you to look at the data at a low level — something that most people are unprepared to do and that is time-consuming even if you do have the required skills. Often, your first clue that data is missing is the preposterous answers that your questions elicit from the algorithm and associated dataset. When the algorithm is the right one to use, the dataset must be at fault. Here are some issues to consider:

- **Essential data missing:** A problem can occur when the data collection process lacks all the data necessary to answer a particular question. Sometimes you're better off to drop a fact than to use a considerably damaged fact.

- **Some data missing:** Less damaged fields can have data missing in one of two ways — randomly or sequentially:
  - **Randomly missing data is often the result of human or sensor error.** Fixing randomly missing data is easiest. You can use a simple median or average value as a replacement. No, the dataset isn't completely accurate, but it will likely work well enough to obtain a reasonable answer.
  - **Sequentially missing data occurs during some type of generalized failure.** Fixing sequentially missing data is significantly harder, if not impossible, because you lack any surrounding data on which to base any sort of guess. If you can find the cause of the missing data, you can sometimes reconstruct it.

Considering data misalignments

Data might exist for each of the data records in a dataset, but it might not align with other data in other datasets you own. For example, the numeric data in a field in one dataset might be a floating-point type (with decimal point), but an integer type in another dataset. Before you can combine the two datasets, the fields must contain the same type of data.

All sorts of other kinds of misalignment can occur. For example, date fields are notorious for being formatted in various ways. To compare dates, the data formats must be the same. However, dates are also insidious in their propensity for looking the same but not being the same. For example, dates in one dataset might use Greenwich Mean Time (GMT) as a basis, whereas the dates in another dataset might use some other time zone. Before you can compare the times, you must align them to the same time zone. It can become even weirder when dates in one dataset come from a location that uses daylight saving time (DST) but dates from another location don't.

Even when the data types and format are the same, other data misalignments can occur. For example, the fields in one dataset may not match the fields in the other dataset. In some cases, these differences are easy to correct. One dataset may treat first and last names as a single field, while another dataset might use separate fields for first and last names. The answer is to change all datasets to use a single field or to change them all to use separate fields for first and last names. Unfortunately, many misalignments in data content are harder to figure out. In fact, it's entirely possi-

ble that you might be unable to figure them out. However, before you give up, consider these potential solutions to the problem:

- Calculate the missing data from other data you can access.
- Locate the missing data in another dataset.
- Combine datasets to create a whole that provides consistent fields.
- Collect additional data from various sources to fill in the missing data.
- Redefine your question so that you no longer need the missing data.

Separating useful data from other data

Some organizations' leaders are of the opinion that they can never have too much data, but an excess of data becomes as much a problem as not enough. To solve problems efficiently, an AI requires just enough data. Defining the question that you want to answer concisely and clearly helps, as does using the correct algorithm (or algorithm ensemble). Of course, the major problems with having too much data are that finding the solution (after wading through all that extra data) takes longer, and sometimes you get confusing results because you can't see the forest for the trees.



WARNING As part of creating the dataset you need for analysis, you make a copy of the original data rather than modify it. Always keep the original, raw data pure so that you can use it for other analysis later. In addition, creating the right data output for analysis can require a number of tries because you may find that the output doesn't meet your needs. The point is to create a dataset that contains only the data needed for analysis, but keep in mind that the data may need specific kinds of pruning to ensure the desired output.

Considering the Five Mistruths in Data

Humans are used to seeing data for what it is, in many cases: an opinion. In fact, in some cases, people skew data to the point where it becomes useless, a *mistruth.* A computer can't tell the difference between truthful and untruthful data — all it sees is data. One issue that makes it difficult,

if not impossible, to create an AI that actually thinks like a human is that humans can work with mistruths, and computers can't. The best you can hope to achieve is to see the errant data as outliers and then filter it out, but that technique doesn't necessarily solve the problem because a human would still use the data and attempt to determine a truth based on the mistruths that are there.



WARNING A common thought about creating less contaminated datasets is that, instead of allowing humans to enter the data, collecting the data via sensors or other means should be possible. Unfortunately, sensors and other mechanical input methodologies reflect the goals of their human inventors and the limits of what the particular technology is able to detect. Consequently, even machine- or sensor-derived data is also subject to generating mistruths that are quite difficult for an AI to detect and overcome.

The following sections use a car accident as the main example to illustrate five types of mistruths that can appear in data. The concepts that the accident is trying to portray may not always appear in data, and they may appear in different ways than discussed. The fact remains that you normally need to deal with these sorts of issues when viewing data.

Commission

Mistruths of *commission* are those that reflect an outright attempt to substitute truthful information for untruthful information. For example, when filling out an accident report, someone could state that the sun momentarily blinded them, making it impossible to see someone they hit. In reality, perhaps the person was distracted by something else or wasn't actually thinking about driving (possibly considering a nice dinner). If no one can disprove this theory, the person might get by with a lesser charge. However, the point is that the data would also be contaminated. The effect is that now an insurance company would base premiums on errant data.



REMEMBER Although it would seem that mistruths of commission are completely avoidable, often they aren't. Humans tell "little white lies" to save others from embarrassment or to deal with an issue with the least amount of personal effort. Sometimes a mistruth of commission is based on errant input or hearsay. In fact, the sources of errors of commission are so many that it is truly difficult to come up with a scenario where someone could avoid them entirely. Regardless, mistruths of commission are one type of mistruth that someone can avoid more often than not.

Omission

Mistruths of *omission* are those in which a person tells the truth in every stated fact but leaves out an important fact that would change the perception of an incident as a whole. Thinking again about the accident report, say that your car strikes a deer, causing significant damage to your car. You truthfully say that the road was wet; it was near twilight, so the light wasn't as good as it could be; you were a little late in pressing on the brake; and the deer simply darted out from a thicket at the side of the road. The conclusion would be that the incident is simply an accident.

However, you left out an important fact: You were texting at the time. If law enforcement knew about the texting, it would change the reason for the accident to inattentive driving. You might be fined, and the insurance adjuster would use a different reason when entering the incident into the database. As with the mistruth of commission, the resulting errant data would change how the insurance company adjusts premiums.



REMEMBER Avoiding mistruths of omission is nearly impossible. Yes, people can purposely leave facts out of a report, but it's just as likely that they'll simply fail to include all the facts. After all, most people are quite rattled after an accident, so they can easily lose focus and report only those truths that leave the most significant impression. Even if a person later

remembers additional details and reports them, the database is unlikely to ever contain a full set of truths.

Perspective

Mistruths of *perspective* occur when multiple parties view an incident from multiple vantage points. For example, in considering an accident involving a struck pedestrian, the person driving the car, the person getting hit by the car, and a bystander who witnessed the event would all have different perspectives. An officer taking reports from each person would understandably glean different facts from each one, even assuming that each person tells the truth as each knows it. In fact, experience shows that this is almost always the case, and the info that the officer submits as a report is the middle ground of what each of those involved states, augmented by personal experience. In other words, the report will be close to the truth, but not close enough for an AI.

When dealing with perspective, consider vantage point. The driver of the car can see the dashboard and knows the car's condition at the time of the accident. This is information that the other two parties lack. Likewise, the person getting hit by the car has the best vantage point for seeing the driver's facial expression (intent). The bystander might be in the best position to see whether the driver made an attempt to stop, and assess issues such as whether the driver tried to swerve. Each party will have to make a report based on seen data without the benefit of hidden data.



WARNING Perspective is perhaps the most dangerous of the mistruths because anyone who tries to derive the truth in this scenario ends up, at best, with an average of the various stories, which will never be fully correct. A human viewing the information can rely on intuition and instinct to potentially obtain a better approximation of the truth, but an AI will always use just the average, which means that the AI is always at a significant disadvantage. Unfortunately, avoiding mistruths of perspective is impossible because no matter how many witnesses you have to the event,

the best you can hope to achieve is an approximation of the truth, not the actual truth.

Think about this other scenario that involves perception: You're a deaf person in 1927. Each week, you go to the theater to view a silent film, and for an hour or more, you feel like everyone else. You can experience the movie in the same way everyone else does; there are no differences. In October of that year, you see a sign saying that the theater is upgrading to support a sound system so that it can display *talkies* — films with a soundtrack.

The sign says that talkies are the best thing ever, and almost everyone seems to agree, except for you, the deaf person, who is now made to feel like a second-class citizen — different from everyone else and even pretty much excluded from the theater. In the deaf person's eyes (from their perspective), that sign is a mistruth; adding a sound system is the worst possible thing, not the best possible thing. The point is that what seems to be generally true isn't actually true for everyone. The idea of a general truth — one that is true for everyone — is a myth. It doesn't exist.

Bias

Mistruths of *bias* occur when someone is capable of seeing the truth but because of personal concerns or beliefs is unable to actually see it. For example, when thinking about an accident, a driver might focus attention so completely on the middle of the road that the deer at the edge of the road becomes invisible. Consequently, the driver has no time to react when the deer suddenly decides to bolt into the middle of the road in an effort to cross.



REMEMBER It is known that faster speed blurs peripheral vision so the deer does become harder to see at higher speeds.

A problem with bias is that it can be incredibly hard to categorize. For example, a driver who fails to see the deer can have a genuine accident, meaning that the deer was hidden from view by shrubbery. However, the

driver might also be guilty of inattentive driving because of incorrect focus. The driver might also experience a momentary distraction. In short, the fact that the driver didn't see the deer isn't the question; instead, it's a matter of *why* the driver didn't see the deer. In many cases, confirming the source of bias becomes important when creating an algorithm designed to avoid a biased source.



REMEMBER Theoretically, avoiding mistruths of bias is always possible. In reality, however, all humans have biases of various types, and those biases will always result in mistruths that skew datasets. Just persuading someone to actually look and then see something — to have it register in the person's brain — is a difficult task. Humans rely on filters to avoid information overload, and these filters are also a source of bias because they prevent people from actually seeing things.

Frame of reference

Of the five mistruths, frame of reference need not be the result of any sort of error, but one of understanding. A *frame-of-reference* mistruth occurs when one party describes something, such as an event like an accident, and because a second party lacks experience with the event, the details become muddled or completely misunderstood. Comedy routines abound that rely on frame-of-reference errors. One famous example is from Abbott and Costello, *Who's On First*, which you can find on `YouTube.com`. Getting one person to understand what a second person is saying can be impossible when the first person lacks experiential knowledge — the frame of reference.

Another frame-of-reference mistruth example occurs when one party can't possibly understand the other. For example, a sailor experiences a storm at sea. Perhaps it's a monsoon, but assume for a moment that the storm is substantial — perhaps life-threatening. Even with the use of videos, interviews, and a simulator, the experience of being at sea in a life-threatening storm would be impossible to convey to someone who

hasn't experienced such a storm firsthand; that person has no frame of reference.



REMEMBER The best way to avoid frame-of-reference mistruths is to ensure that all parties involved can develop similar frames of reference. To accomplish this task, the various parties require similar experiential knowledge to ensure the accurate transfer of data from one person to another. However, when working with a dataset, which is necessarily recorded static data, frame-of-reference errors will still occur when the prospective viewer lacks the required experiential knowledge.

An AI will always experience frame-of-reference issues because it necessarily lacks the ability to create an experience. A data bank of acquired knowledge isn't quite the same thing. The data bank would contain facts, but experience is based on not only facts but also conclusions that current technology is unable to duplicate.

Defining the Limits of Data Acquisition

If you get the feeling that everyone is acquiring your data without thought or reason, you're right. In fact, organizations collect, categorize, and store everyone's data — seemingly without goal or intent.

Data acquisition has become a narcotic for organizations worldwide, and some of their leaders seem to think that the organization that collects the most somehow wins a prize. However, data acquisition, in and of itself, accomplishes nothing. The book *The Hitchhiker's Guide to the Galaxy,* by Douglas Adams, illustrates this problem clearly. In this book, a race of supercreatures builds an immense computer to calculate the meaning of "life, the universe, and everything." The answer of 42 doesn't really solve anything, so some of the creatures complain that the collection, categorization, and analysis of all the data used for the answer hasn't produced a usable result. The computer — a sentient one, no less — tells the people receiving the answer that the answer is indeed correct, but they need to know the question in order for the answer to make sense. Data acquisi-

tion can occur in unlimited amounts, but figuring out the right questions to ask can be daunting, if not impossible.



REMEMBER The main problem that any organization needs to address with regard to data acquisition is which questions to ask and why the questions are important. Tailoring data acquisition to answer the questions you need answered matters. For example, if you're running a shop in town, you might need answers to questions like this:

- How many people walk in front of the store each day?
- How many of those people stop to look in the window?
- How long do they look?
- What time of day are they looking?
- Do certain displays tend to produce better results?
- Which of these displays cause people to enter the store and shop?

The list could go on, but the idea is that creating a list of questions that address specific business needs is essential. After you create a list, you must verify that each of the questions is indeed important — that is, addresses a need — and then ascertain what sorts of information you need to answer the question.



WARNING Of course, trying to collect all this data by hand would be impossible, which is where automation comes into play. Seemingly, automation would produce reliable, repeatable, and consistent data input. However, many factors in automating data acquisition can produce data that isn't particularly useful. For example, consider these issues:

- Sensors can collect only the data they're designed to collect, so you might miss data when the sensors used aren't designed for the purpose.
- People create errant data in various ways (see the "Considering the Five Mistruths in Data" section, earlier in this chapter, for details), which means that data you receive might be false.
- Data can become skewed when the conditions for collecting it are incorrectly defined.

- Interpreting data incorrectly means that the outputs from AI will also be incorrect.
- Converting a real-world question into an algorithm that the computer can understand is an error-prone process.

Many other issues (enough to fill a book) need to be considered. When you combine poorly collected, ill-formed data with algorithms that don't actually answer your questions, you get output that may actually lead your business in the wrong direction, which is why AI is often blamed for inconsistent or unreliable results. Asking the right question, obtaining the correct data, performing the right processing, and then correctly analyzing the data are all required in order to make data acquisition the kind of tool you can rely on.

Considering Data Security Issues

This section discusses data security from the perspective of protecting data integrity rather than keeping someone from stealing it or guarding privacy. Securing data doesn't mean placing it in a vault — assuming that doing so is even possible with data today. Data is useful only when it's accessible. Of course, the need to make data accessible means taking a risk that someone will do something you don't want done with the data. The following sections discuss a few data security issues you need to consider.

Understanding purposefully biased data

Bias appears in nearly every dataset available today, even custom-created datasets. The dataset is often biased because the collection methods are biased, the analysis methods are biased, and the data itself is biased. You often see articles online with titles like "8 Types of Bias in Data Analysis and How to Avoid Them," which means that people recognize the existence of bias and want to mitigate it as much as possible. However, sometimes you find that the opposite is true: The people using the dataset purposely bias it in some manner. Here are some areas in which data becomes purposely biased:

- **Political:** Political maneuvering can become the source of data bias. Two groups with opposing opinions will use the same dataset and obtain two completely dif-

ferent outcomes that support their particular perspective. At issue are the records selected and the dataset features used to create an outcome. In other cases, a group will resort to techniques like using bogus respondents in polls (see "Assessing the Risks to Online Polls from Bogus Respondents" at `pewresearch.org` for details).

- **Medical:** When medical groups advertise for people to participate in trials of medications, procedures, and other needs, the group they get often doesn't represent the population as a whole, so the data is biased.
- **Legal:** The use of COMPAS software (it's short for Correctional Offender Management Profiling for Alternative Sanctions) to predict the potential for *recidivism* (the tendency of a convicted criminal to reoffend) is another example of data and algorithm bias, as explained in "Injustice Ex Machina: Predictive Algorithms in Criminal Sentencing," at `uclalawreview.org`. The article points out so many flaws with COMPAS that the best idea might be to start from scratch because the software is destroying people's lives at an incredible rate.
- **Hiring:** The use of datasets and well-rounded algorithms supposedly reduces the risk of bias in hiring and promoting individuals within an organization. According to "All the Ways Hiring Algorithms Can Introduce Bias" at `hbr.org`, the opposite is too often true. The datasets become an amplification of biased hiring practices within an organization or within society as a whole.
- **Other:** Anytime a dataset and its associated algorithms become influenced by bias, the outcome is less than ideal. The term *machine learning fairness* presents the idea that the outcome of any analysis should correctly represent the actual conditions within. If the outcome of an analysis doesn't match the result received afterward, the analysis is flawed and the data usually receives a lion's share of the blame.

Dealing with data-source corruption

Even if people don't cherry-pick data or use data sources that fail to reflect the actual conditions in the world, as described in the previous section, data sources can become corrupt. For example, when seeing product reviews on a website, you can't be certain that

- Real people created reviews.
- Some people haven't voted more than once.
- The person wasn't simply having an exceptionally bad (or less likely, good) day.
- The person actually used the product and has no ulterior motive, for example, if they sell the product or are a competitor.
- The reviews reflect a fair segment of society.

In fact, the reviews are likely so biased and corrupt that believing them at all becomes nearly impossible. Unfortunately, data-source corruption comes from many other sources:

- A sensor might be bad, producing erroneous results.
- A virus attack might cause data errors.
- The database or other software contains a flaw.
- Humans enter the data incorrectly into the database.
- Acts of nature, such as lightning, cause momentary glitches in data collection.

You can rely on a number of approaches to deal with all sorts of data corruption. Storing data in the cloud tends to reduce problems associated with hardware, weather, or other issues that cause data loss. Ensuring that you have procedures and training in place, plus constant monitoring, can help reduce human errors. Active administrator participation and the use of firewalls can reduce other sorts of data-source corruption.



REMEMBER All these measures reflect what you can do locally. When performing screen scraping and other techniques to obtain data from online sources, data scientists must employ other measures to ensure that the data remains pure. Vouching for an online source isn't possible unless the source is vetted every time it's used.

Handling botnets

Botnets are coordinated groups of computers that focus on performing specific tasks, most of them nefarious. This short section focuses on botnets that feed a dataset erroneous data or take over accounts to modify the account information in certain ways. Whatever means is used, whatever the intent, botnets generally corrupt or bias data in ways that cause any kind of analysis to fail.

Botnets represent a significant concern in AI for several reasons, primarily due to their evolving capabilities and the scale and speed of attacks they can launch. Here are three key reasons that botnets are a particular concern:

- **AI-enhanced botnets:** With advances in AI and machine learning, botnets have become more sophisticated, capable of automating and rapidly expanding cyberattacks. AI can enable botnets to analyze network behavior, adapt attack patterns to bypass cyberdefenses, and execute attacks with increased efficiency and stealth. This adaptability makes AI-enhanced botnets formidable adversaries, capable of evading traditional detection mechanisms and launching potent and destructive attacks.
- **Scale and magnitude of attacks:** AI-powered botnets can harness the computational power of numerous compromised devices, including the internet of things (IoT), creating massive bot armies. With these vast networks, AI-powered botnets can launch coordinated attacks that overwhelm even the most robust security infrastructures. This scale and magnitude of attacks pose significant challenges to cyberdefenses, requiring advanced detection and mitigation strategies.
- **Rapid attack development:** Unlike traditional botnets that require manual programming for specific targets, AI-powered counterparts leverage algorithms to analyze and exploit vulnerabilities quickly. This enables them to develop new attack techniques at an alarming pace, keeping defenders on their toes and making it difficult to predict and prevent attacks.



WARNING The continuous mutation of botnets to exploit vulnerabilities and security flaws makes prevention and mitigation challenging. Botnet operators use a variety of IP addresses and devices in their attacks, complicating the task of screening out bad requests and confidently allowing access to valid requests. The proliferation of IP-addressable IoT devices has expanded the potential for botnets to spread and launch attacks.